

On the orthogonality penalty in spectrally parameterised physics-informed neural networks: factor geometry and transfer to shock-dominated flow

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Abstract

Wang et al. [16] improve physics-informed neural network accuracy by factorising each hidden weight at initialisation, training the left factor and the singular values in place of the weight, and penalising departures of the left factor from orthogonality with a hand-tuned weight w_U . Their account assigns the gain to control over the weight spectrum. We reproduce their advection case over 20 seeds and then measure the trained factors, which their paper does not report. At the published $w_U = 1/35000$ the trained left factor is not close to orthogonal — $\|U^\top U - I\|_2$ has median 0.98, indicating a collapsed eigendirection — and the trained singular values differ from the effective weight’s actual singular values by up to a factor of 13. Constraining the factor to be exactly orthogonal, which removes w_U entirely, reaches only 1.7×10^{-2} median relative error against the penalty’s 8.4×10^{-3} ; removing the penalty from the same parameterisation returns the error to 2.1×10^{-1} , worse than an unconstrained network’s 8.4×10^{-2} . Reading the term as a bound on the factor’s scale predicts that a weaker bound should also work; we test that by pinning only $\sigma_{\max}(U)$, and it is the worst regime of the six at 3.7×10^{-1} median, so what the penalty supplies is a pull of the whole spectrum toward 1 rather than a bound on scale. Transferring the comparison to two-dimensional shallow-water dam breaks, where the solution carries a shock and the bounded-gradient hypothesis of the source paper’s Theorem 2.1 fails, no spectral regime separates from an unmodified network by more than about 10% in L^1 depth error on the three shock cases, and w_U can be moved three decades without measurable effect. On the one smooth initial condition in the set, exact orthogonality and spectrum-only training are a factor of 2.5 worse than an unmodified network. Establishing this required checking our finite-volume reference against an independently implemented solver family, which disagrees with it by more than the differences under test on two of three benchmarks — a floor that comparisons of this kind do not usually report.

1 Introduction

A physics-informed neural network represents the solution of a PDE by a network and trains it by minimising the residual of the equation at sampled collocation points, together with terms for the initial and boundary data [9]. The appeal is that the same machinery handles forward and inverse problems, irregular geometries and partial data without a mesh. The difficulty, visible from the earliest applications, is that training often stalls on solutions with steep gradients or high-frequency

content — the regime where classical solvers are also hardest, and therefore the regime that would most justify the method.

Several distinct explanations have been offered for that failure and each comes with its own remedy. Networks trained by gradient descent fit low frequencies before high ones [8], which motivates input encodings that raise the frequency content the network can express cheaply: random Fourier features [12] and periodic activations [10] are the standard instruments, and adaptive activation slopes are a lighter variant [4]. A second account locates the problem in the loss rather than the architecture: residual and boundary terms have gradients of very different magnitude, so one term dominates and training converges to a solution that satisfies the wrong constraint. That reading has produced gradient-based rebalancing [14], neural-tangent-kernel weighting [15], self-adaptive pointwise weights [6] and augmented-Lagrangian treatments of the boundary conditions [11]. A third account, the one this paper is about, looks at the weights themselves.

Wang et al. [16] propose a diagnostic and a remedy built on a quantity they call the condition number of the network, $\kappa(f) = \|J(f)\|_F$. The name invites confusion — this is an absolute condition number, an H^1 seminorm of the gradient magnitude, and not the $\sigma_{\max}/\sigma_{\min}$ of numerical linear algebra — but the construction is concrete. Each hidden weight is factorised once at initialisation as $W_j = U_j S_j V_j^\top$; V_j is frozen; U_j and $\text{diag}(S_j)$ are trained in place of W_j ; and the loss gains

$$w_U \sum_j D_j^2, \quad D_j = \|U_j^\top U_j - I\|_2 \quad (1)$$

so that U_j is pushed toward orthogonality and $\text{diag}(S_j)$ therefore carries the singular values of W_j . The stated mechanism is that the optimiser gains direct control of each layer’s spectrum, which sets the Lipschitz scale the network can express, and that this is what helps on steep targets. Across five PDEs the method improves on a vanilla network by roughly an order of magnitude.

Their conclusion lists three open problems, and they are the starting point here. First, w_U is tuned per equation across two and a half decades, from 1/600 to 1/200000, and their Section 4.6.3 reports that NTK- and DB-PINN-style adaptive weighting diverges on this particular term because the penalty begins at exactly zero, so the adapted weight grows without bound. Second, the penalty costs 1.28 to 1.83 times vanilla training time across their width and collocation sweeps. Third, all five validation problems are, in their own words, controlled synthetic examples, and all five are smooth.

The third point has a consequence the paper does not draw. Every hypothesis of their Theorem 2.1 requires $u \in H^1$ with $\|J(u)\|_F$ bounded. A dam-break solution carries a shock, so $\|J(u)\|_F$ is unbounded and $u \notin H^1$: the theorem is formally inapplicable to the whole problem class, and the diagnostic that motivates the method is a quantity that does not exist there. The authors come close to saying so in another context, noting that the condition is violated once training fully collapses. Whether the *method* still helps when its *theory* does not apply is an empirical question, and it is the one that makes shock-dominated flow the informative test rather than merely a harder one.

The apparatus for that test comes from a companion study by an overlapping author group, submitted to *Computers & Fluids* [17]. That study compares classical finite-volume schemes against several PINN formulations on six two-dimensional dam-break initial conditions under one set of conventions, and reports that a physics-only, finite-volume-informed network fails on the short cases where a strong-form network merely does poorly. We reuse its benchmark definitions, its conventions, and its differentiable HLLC solver, and we inherit its finding about the finite-volume-informed baseline rather than rederiving it. Two things follow. The comparison here is a controlled substitution — only the weight parameterisation changes between entries, on fixed benchmarks and

a fixed evaluation protocol — and the reference solver output produced for that study, computed by independent implementations of HLL, Lax–Wendroff and MUSCL–Rusanov, is available to check our own reference against. Section `refsec:refcheck` reports that comparison.

This paper addresses the three limitations in turn and extends the evaluation to the excluded regime. The principal findings are as follows.

- Reproducing the advection case over 20 seeds and then measuring the trained factors — which the source paper does not report — shows that the stated mechanism does not survive training. At the published w_U the trained U is rank-deficient rather than near-orthogonal, and $\text{diag}(S)$ is not the effective weight’s spectrum. The accuracy gain is real; the explanation for it is not (Section 4).
- Removing the penalty by constraining U to be exactly orthogonal is stable but recovers only about half the gain, and replacing it with a bound on U ’s largest singular value — the weaker constraint that reading suggests — is the worst regime we tried (Section 4.3).
- On shock-dominated shallow-water dam breaks no spectral regime separates from an unmodified network by more than about 10% in integrated depth error, and w_U can be moved three decades without measurable effect, so the tuning burden of the first limitation does not arise here (Sections 5, 5.3). On the one smooth case in the set, the constrained regimes are a factor of 2.5 *worse* than an unmodified network.
- The overhead of the second limitation does not reproduce: the penalty costs 3%, while the penalty-free alternative that would replace it costs 42%, so eliminating w_U costs time rather than saving it (Section 5.4).
- Establishing any of this required checking our finite-volume reference against an independently implemented solver family. On two of three benchmarks the two disagree by more than the differences the comparison is trying to resolve (Section 5.5). We report this at length because a single fine-grid run is the usual reference in this literature and its uncertainty is rarely quoted.

2 Related work

Where the failure is located. The three accounts sketched in the introduction — architecture, loss balance, weight structure — are not exclusive, and the same benchmark often improves under all three. That makes attribution hard and is part of why this paper spends its effort on measurement rather than on a new variant. The architectural line treats the network’s frequency response as the binding constraint and modifies the input encoding or the activation [12, 10, 4]; we include a Fourier-feature network as a baseline throughout, because it is the standard remedy for exactly the steep targets [16] opens with and it is absent from that paper’s comparison. A sine activation appears there only as an ablation of their own method, not as a SIREN baseline.

Loss balance and constraint handling. The loss-weighting line rebalances terms whose gradients differ in magnitude [14, 15, 6]. AL-PINNs [11] instead treat the initial and boundary data as constraints and apply an augmented-Lagrangian method, replacing a fixed weight by a multiplier with an update rule. The distinction from the present work is what the constraint is placed on: those methods constrain the *solution* the network represents, whereas the penalty studied here constrains the *weights*. The distinction matters concretely, because it explains the failure [16] report in

their Section 4.6.3: adaptive schemes from this family diverge on their penalty because it begins at exactly zero, a pathology specific to a constraint that is satisfied at initialisation by construction.

Spectral structure in the weights. Spectral normalisation [7] divides each weight by its largest singular value to enforce a 1-Lipschitz map. It acts on the same object as the method studied here, with the opposite intent: it *caps* the gradient magnitude the network can express, whereas the diagnostic motivating cnPINN holds that networks fail on steep targets by expressing too little of it. Our bounded regime (Section 4.3) is close to spectral normalisation in form, and the fact that it is the worst regime we tried is consistent with the two having opposite aims. Exactly-orthogonal weights are available through the matrix-exponential and Cayley parametrisations of Lezcano-Casado and Martínez-Rubio [5], introduced for recurrent networks where the motivation is gradient stability over long sequences; we are not aware of a prior application to PINN training. SVD-PINNs [3] use the same three-factor layer as our spectrum-only regime, freezing both orthogonal factors and retraining the singular values, but for transfer between equations rather than from-scratch training. The construction is theirs; what we add is its behaviour as a from-scratch method, which is poor.

PINNs for hyperbolic problems and shocks. A pointwise PDE residual is not defined at a discontinuity, so the line of work aimed at conservation laws changes the objective or the architecture rather than the weights. Weak and entropy formulations replace the strong residual with one that remains defined for discontinuous solutions and select the physically admissible weak solution [2]; other approaches impose jump conditions through the architecture, concentrate loss weight near detected shocks, or borrow relaxation formulations from the numerical-analysis literature. None of these measures or controls a property of the network’s weights, so the comparison reported here is orthogonal to them rather than competing with them. Composing a spectral parametrisation with a weak-form loss is untested, and we do not test it: our finite-volume-informed baseline is a different composition, in which the residual is that of a discrete conservative update rather than of the differential equation.

Shallow-water dam breaks. PINNs have been applied to the shallow-water equations for flood and dam-break prediction, generally on smooth or mildly discontinuous cases, and hybrids that embed a finite-volume update in the loss are the closest relatives of our physics-only baseline. That literature is surveyed in the companion study of [17], from which our benchmark definitions, conventions ($[0, 100]^2$ m domain, $g = 2$ m/s², Gaussian-hump bed, reflective walls) and differentiable HLLC solver are taken. Two of that study’s results carry directly into this one. Its finding that the finite-volume-informed formulation collapses on short dam-break windows is reproduced here as a baseline rather than reinvestigated (Section 5). Its reference solver output — HLL, Lax–Wendroff and MUSCL–Rusanov runs on a 501^2 node grid for the same initial conditions, produced by an implementation independent of ours — is what makes Section 5.5 possible; without a second implementation there would be no way to put an uncertainty on the reference at all. Reusing a companion study’s apparatus buys a controlled comparison and costs independence, and we flag both.

Reference practice. We found no comparable check in the PINN-for-conservation-laws literature. The standard is to compare against one fine-grid run of one scheme, or against an analytical solution where one exists, and to report neural errors to two or three significant figures without an uncertainty on the reference itself. Section 5.5 shows that on two of our three checkable benchmarks that uncertainty exceeds the differences being reported, which suggests the practice is worth

revisiting beyond this paper.

What we did not find. A search in August 2026 returned no PINN work training weights under an exact orthogonality constraint, no condition-number or gradient-norm mismatch diagnostic for PINNs on hyperbolic problems, and no indexed citation of [16], which had appeared online two months earlier. The searches covered arXiv-indexed material and are not exhaustive — in particular, a method published under different terminology for the same construction would not necessarily surface — so the sweep needs repeating before submission.

3 Method

3.1 The factorised layer

Write a hidden layer’s weight as $W \in \mathbb{R}^{m \times n}$. At initialisation, and only there, it is factorised by singular value decomposition as

$$W_0 = U_0 S_0 V_0^\top, \quad U_0 \in \mathbb{R}^{m \times r}, \quad S_0 = \text{diag}(s_0), \quad V_0 \in \mathbb{R}^{n \times r}, \quad (2)$$

with $r = \min(m, n)$. Thereafter the layer’s effective weight is rebuilt from the stored factors on every forward pass, $W = U \text{diag}(s) V^\top$, and the decomposition is never recomputed during training. Following the source paper, V stays frozen at V_0 , the first and last layers of the network keep ordinary dense weights, and only the square hidden-to-hidden weights are factorised. The parameter count rises by r per factorised layer.

If U were orthogonal, the singular values of W would be exactly $|s|$, and the optimiser would have direct control of each layer’s spectrum. That is the mechanism the method is built on, and Eq. (1) is what is supposed to keep U close enough to orthogonal for it to hold. Whether it does is the question of Section 4.

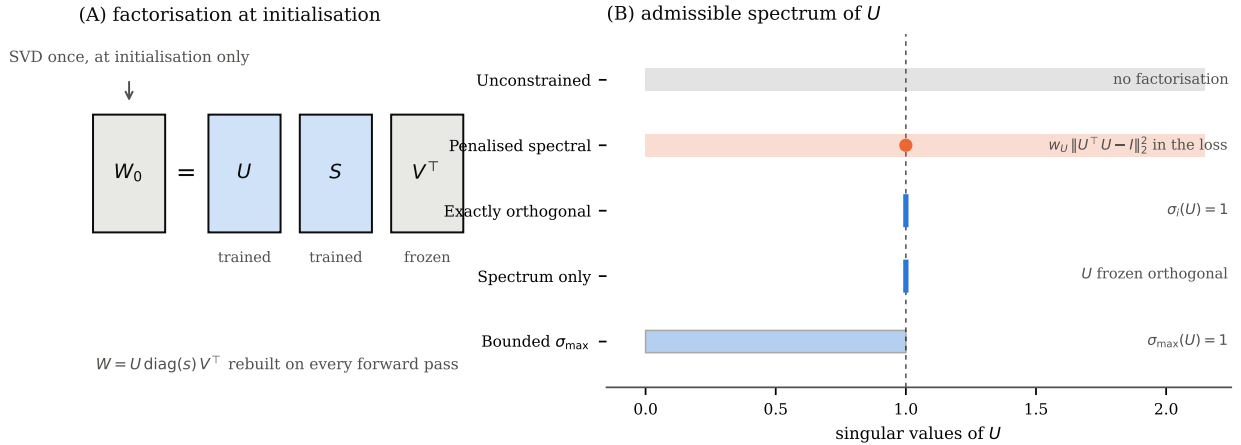


Figure 1: (Color online) (A) Each square hidden weight is factorised once, at initialisation, into $USV^\top$; V is then held fixed and the effective weight is rebuilt from the stored factors on every forward pass. (B) The constraint each parameterisation places on the singular values of U . Exact orthogonality admits the single point $\sigma_i = 1$; the penalised parameterisation admits any spectrum but attaches a cost that grows with the distance from that point; bounding $\sigma_{\max}$ admits the whole interval $[0, 1]$, which includes rank-deficient factors.

3.2 Weight parameterisations compared

Five parameterisations of the hidden layers are compared. They share Eq. (2) and keep $V = V_0$ frozen, and differ in what is trained and in what constrains U . Fig. 1 shows the construction and the admissible spectrum of U in each case; Table 1 summarises them.

In the unconstrained case the weight W is trained directly, without factorisation; this is the vanilla network and serves as the baseline. The penalised parameterisation trains U and s and adds $w_U \sum_j D_j^2$ from Eq. (1) to the loss, which is the method of [16] and carries the single free parameter w_U .

The exactly orthogonal parameterisation confines U to the Stiefel manifold using the matrix-exponential map of Lezcano-Casado and Martínez-Rubio [5], so that $U^\top U = I$ holds to machine precision after every optimiser step and the singular values of W equal $|s|$ exactly. The penalty term is then redundant and w_U does not appear. This parameterisation and the penalised one target the same property by different means, one approximately with a tuned weight and one exactly without, so a comparison between them isolates the contribution of the penalty from the contribution of the factorisation.

In the spectrum-only parameterisation both U and V are frozen at their orthogonal initial values and s alone is trained, giving r trainable parameters per layer in place of mn and exact spectral control by construction. The same three-factor layer is used by SVD-PINNs, where the singular vectors of a trained network are retained and the singular values retrained for a new equation [3]; the construction is not introduced here, and the difference is that training proceeds from scratch.

The fifth parameterisation constrains only the largest singular value of U , rescaling it on every forward pass so that $\sigma_{\max}(U) = 1$ while the remaining singular values stay free in $[0, 1]$. It is therefore strictly weaker than exact orthogonality, which fixes all of them at unity, and it requires no penalty. It is introduced in Section 4.3, where the factor measurements of Section 4.2 motivate it.

Table 1: Weight parameterisations of the hidden layers. All except the unconstrained case use the factorisation of Eq. (2) with V frozen at V_0 ; $r = \min(m, n)$ and $\sigma_i(U)$ denotes the i -th singular value of U .

Parameterisation	Trained	Constraint on U	Free parameter
Unconstrained	W	none	—
Penalised spectral	U, s	$w_U \sum_j D_j^2$ in the loss	w_U
Exactly orthogonal	U, s	$\sigma_i(U) = 1$ for all i	—
Spectrum only	s	U frozen orthogonal	—
Bounded $\sigma_{\max}$	U, s	$\sigma_{\max}(U) = 1$	—

3.3 Implementation

The penalty of Eq. (1) uses the matrix 2-norm, following the source paper. An early version of our code used the Frobenius norm, which upper-bounds it by up to $\sqrt{n}$ and so is up to 128 times stronger at width 128. Both are reported rather than one being discarded: the dam-break matrix of Table 4 was produced with the Frobenius penalty and the sweep of Table 5 with Eq. (1), and Section 5.3 compares them on the same benchmarks. Only the penalised entry reads that norm, so no other row is affected by the change.

The advection case follows the source paper’s protocol: L-BFGS with a strong-Wolfe line search, 3000 iterations. The shallow-water cases use Adam at 10^{-3} for 20000 iterations, since L-BFGS on a

two-dimensional time-dependent problem with 20000 resampled collocation points is not practical. One regime forces a departure from this: the singular-value box of Section 4.3 cannot be trained by L-BFGS at all, for reasons given there, so where it appears the optimiser is stated explicitly and its numbers are not compared across protocols.

Random seeds are set before model construction rather than before the training loop, so initial weights — and, where used, the Fourier feature matrix — are reproducible from the seed alone. Every reported cell is five seeds on the shallow-water benchmarks and twenty on advection.

4 The advection benchmark

4.1 Reproduction of the published comparison

Table 2 repeats the source paper’s Section 4.1 advection setup: a 3×25 tanh network, Xavier initialisation, 100 boundary and 500 interior points by Latin hypercube sampling, L-BFGS to 3000 iterations, $w_U = 1/35000$. We report 20 seeds. Distributions in this problem are heavy-tailed — the unconstrained network’s worst seed is 53 times its best — so the median is the more stable summary and both are given.

Table 2: Relative L^2 error on the advection case over 20 seeds. The first two rows are the comparison the source paper reports (their Table 5, tanh activation); the last four are regimes it does not test. The two published cells are reproduced within their reported spread.

Regime	Mean $\pm$ s.d.	Median	Wang et al. Table 5
Unconstrained	$(1.37 \pm 1.4) \times 10^{-1}$	8.4×10^{-2}	$(1.13 \pm 1.67) \times 10^{-1}$
Penalised, $w_U = 1/35000$	$(1.46 \pm 1.5) \times 10^{-2}$	8.4×10^{-3}	$(5.64 \pm 3.86) \times 10^{-3}$
Exactly orthogonal	$(3.31 \pm 3.9) \times 10^{-2}$	1.7×10^{-2}	—
Spectrum only	$(3.29 \pm 5.5) \times 10^{-1}$	1.2×10^{-1}	—
Bounded $\sigma_{\max}(U)$	$(4.68 \pm 5.1) \times 10^{-1}$	3.7×10^{-1}	—
Same parameterisation, $w_U = 0$	$(3.37 \pm 3.5) \times 10^{-1}$	2.1×10^{-1}	—

The two published cells are reproduced within their reported spread, which establishes that the setup is faithful. Three observations follow from the remaining rows. Exact orthogonality, which eliminates w_U , costs approximately a factor of two in median error relative to the tuned penalty. Training the spectrum alone is less accurate than an unconstrained network and diverges on 2 of 20 seeds, indicating that the trained directions rather than the trained spectrum carry the accuracy gain; this is already difficult to reconcile with a spectral account of the mechanism. Removing the penalty while retaining the parameterisation gives the second largest median error of the six, ahead only of the bounded parameterisation of Section 4.3.

4.2 Factor geometry after training

The source paper reports errors, not factor geometry. We measure, per factorised layer of trained networks over 3 seeds, the orthogonality defect $D = \|U^\top U - I\|_2$ and the largest relative gap between the trained $|s|$ and the singular values of the effective weight $USV^\top$ (Table 3).

At the published weight $D \approx 1$, which for a defect measured in the 2-norm means an eigendirection of $U^\top U$ has collapsed: the trained factor is rank-deficient, not near-orthogonal. The trained s is not the effective weight’s spectrum. Both premises of the stated mechanism fail, while the method’s accuracy advantage holds.

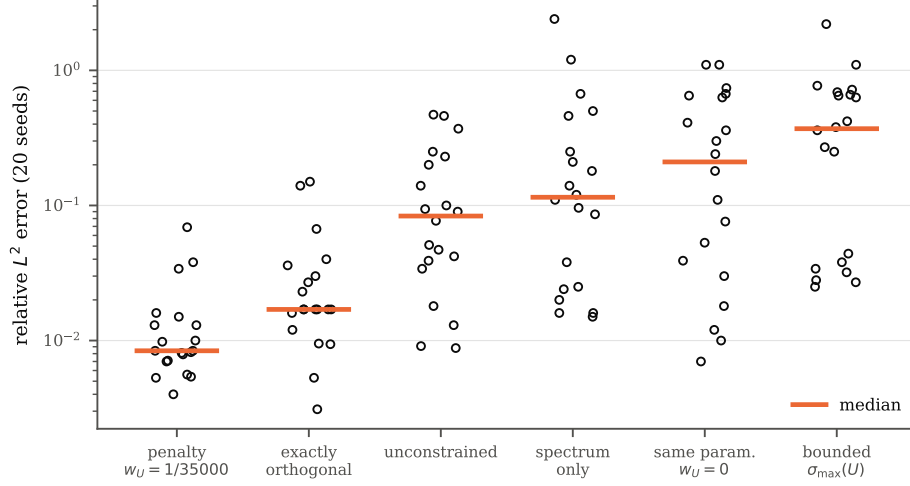


Figure 2: Relative L^2 error on the advection equation for each of the six weight parameterisations, 20 seeds each, on a logarithmic scale and ordered by median. Horizontal bars mark medians. The distributions are heavy-tailed and overlap substantially, so medians are reported alongside means throughout. Bounding $\sigma_{\max}(U)$, the weaker constraint of Section 4.3, gives the largest errors of the six.

Table 3: Factor geometry after training, 3 seeds, median over layers. If the described mechanism held, D would be near zero and the two spectra would coincide. The bounded regime is discussed in Section 4.3; note that it matches the penalised regime’s D while missing the spectrum by two further orders of magnitude.

Setting	Median error	Median D	$ s $ vs. true singular values
$w_U = 1/35000$	7.9×10^{-3}	0.98	up to $13\times$ apart
$w_U = 0$	3.05×10^{-1}	63 (max 177)	up to $31\times$ apart
Bounded $\sigma_{\max}(U)$	3.8×10^{-1}	1.00	up to $1290\times$ apart

Ordering the regimes by median error gives unconstrained-parameterised 2.1×10^{-1} , spectrum-only 1.2×10^{-1} , unconstrained 8.4×10^{-2} , exactly orthogonal 1.7×10^{-2} , weakly penalised 8.4×10^{-3} . The penalty is doing something the method needs: removing it lets the defect grow by two orders of magnitude and destroys the gain, while enforcing the constraint it nominally approximates recovers only about half of it. Exact orthogonality is a stronger constraint than the problem wants. The natural reading is that the term bounds the scale of U rather than orthogonalising it, and Section 4.3 tests that reading directly — it does not survive.

This reframes the first stated limitation: the open question is not how to adapt w_U but which constraint class the term belongs to. The ordering above suggested that a bound weaker than orthogonality — a cap on U ’s largest singular value, or a box on all of them — should recover the gain without a tuned penalty. The first of these is evaluated below.

4.3 A weaker constraint on the left factor

The bounded regime rescales U on every forward pass so that $\sigma_{\max}(U) = 1$, leaving the remaining singular values free in $[0, 1]$. It is strictly weaker than orthogonality, which fixes all of them at 1, and it admits the rank-deficient factor the penalty was measured to produce. It carries no penalty term and no w_U . The scale it removes is not lost to the layer, since $W = U \text{diag}(s) V^\top$ and s is trained.

Over 20 seeds it reaches $(4.68 \pm 5.1) \times 10^{-1}$ mean and 3.7×10^{-1} median relative error: the worst of the six regimes, below even the unpenalised parameterisation. The prediction is refuted.

Measuring the bounded networks’ factors explains the failure and corrects our earlier reading (last row of Table 3). The bounded regime reaches $D = 1.00$, essentially the same defect as the penalised regime’s 0.98 — so on the statistic we used to characterise the penalty, the two are indistinguishable. They are not indistinguishable in behaviour: the gap between the trained $|s|$ and the effective weight’s singular values is $13\times$ under the penalty and $1290\times$ under the cap, and the bounded networks drive their smallest trained singular value to 0 exactly.

The reason is that $D = \max_i |\sigma_i(U)^2 - 1|$ saturates as soon as a single direction of U collapses, so it cannot distinguish one collapsed direction from many. Capping $\sigma_{\max}(U)$ reproduces the value of D while leaving the rest of the spectrum free to collapse, and the effective weight then has nothing to do with the spectrum s nominally controls. The conclusion is narrower than the one we drew from the accuracy ordering alone: what the penalty supplies is not a bound on the factor’s scale — a bound on scale is exactly what the cap provides, and it fails — and D is too coarse a summary to say what it is instead. Section 4’s reading was underdetermined by its own evidence.

The other half of the prediction is a box holding every singular value of U in $[1 - \varepsilon, 1 + \varepsilon]$: the constraint class that matches the measured behaviour, with ε a geometric knob that recovers exact orthogonality at $\varepsilon = 0$ and the unconstrained parameterisation as ε grows, in place of a loss weight on an unnormalised penalty. We implemented it as a projection, which proved untrainable under the present protocol. The projection itself is exact, in that trained networks have every singular value inside the box to five decimal places, but its Jacobian is ill-conditioned precisely where the layer operates: differentiating through the singular value decomposition introduces $1/(\sigma_i^2 - \sigma_j^2)$ terms, and the box holds every σ_i near 1. The exact gradient produced non-finite values within a few hundred iterations. A straight-through estimator is stable and keeps the constraint, but its gradient disagrees with the function value, and L-BFGS with a strong-Wolfe line search then makes almost no progress: over 300 iterations the loss falls from 2.91×10^{-1} to 2.76×10^{-1} , against 1.82×10^{-1} for the exactly-orthogonal regime from the same initialisation. This is therefore an open implementation problem, and does not constitute evidence about the constraint class itself.

The difficulty is itself relevant to the paper’s subject. A soft penalty is differentiable everywhere

and costs one matrix norm per layer; the hard constraint it nominally approximates is a projection whose exact derivative is singular on the set where the constraint binds. That asymmetry is a practical reason for the source paper’s design that its own discussion does not give, and it is a better argument for the penalty than the mechanism the paper offers. Whether a first-order optimiser tolerates the straight-through estimator well enough to make the box trainable remains untested.

The evidence here is one smooth one-dimensional problem, one architecture and one optimiser. Whether the ordering survives on a shock is a separate question, which is what Section 5 asks.

5 Shock-dominated shallow-water benchmarks

5.1 Benchmarks and evaluation protocol

The benchmarks are two-dimensional dam breaks over a Gaussian-hump bed at the conventions of the comparative study of Zahra et al. [17]: domain $[0, 100]^2$ m, $g = 2$ m/s², reflective walls on all four sides, $t_{\text{end}} = 2$ s. We use a step release, a circular column over a wet bed, the same column over a dry bed, and a smooth Gaussian mound, all with Manning $n = 0$. A fifth case, three submerged humps with Manning friction over 20 s, is included and then set aside for the reason given below. Errors are L^1 and L^2 of depth at the final output time on a shared 128^2 evaluation grid, against a reference computed in-process with an HLLC MUSCL–van Leer finite-volume scheme [13] at 512^2 with hydrostatic reconstruction for well-balancing [1]. Baselines are an unmodified network, a Fourier-feature network — the standard remedy for high-frequency targets, absent from the source paper’s comparison — and a finite-volume-informed network. Five seeds per cell.

Every strong-form entry is a 6-layer, 128-unit tanh network mapping (x, y, t) to primitive variables (h, u, v) , trained by Adam at 10^{-3} for 20000 iterations with gradient clipping at 1.0, on 20000 collocation points resampled each step, with the initial condition imposed as a penalty at every cell centre weighted 10 against the residual. Following the source paper, the input and output layers stay dense and only the square hidden weights are factorised. The Fourier-feature baseline uses 32 frequencies. The finite-volume-informed entry predicts cell averages on the solver grid at 41 collocation times and is trained against the residual of the discrete SSP-RK2 update.

5.2 Accuracy across benchmarks

Table 4 gives all five benchmarks, five seeds each. On the three shock-dominated cases no spectral regime separates from an unmodified network by more than about 10% in L^1 depth: on the step release every entry but the finite-volume-informed one falls between 1.03 and 1.13×10^3 , inside the unconstrained baseline’s own seed spread. The unconstrained network sits a factor of 4.1 to 4.4 above a classical scheme at the same resolution.

Fig. 3 resolves that summary on the circular dam break. The reference at $t = 2$ s consists of an outward-travelling bore near $x = 22$ m and $x = 78$ m, a rarefaction behind it, and a plateau whose crest carries a shallow depression left by the Gaussian bed. Panels (B)–(D) show that all three parameterisations reproduce the gross structure; panels (E) and (F) show where they do not. Two error modes dominate and are common to all three. The bore is displaced and smeared, producing deviations of -2 m at the two front positions against -1.0 m for a classical scheme at the evaluation resolution. Behind it the networks return a flat crest near 10.2 m where the reference has a depression near 9.6 m, giving a broad $+1.3$ m deviation across $40 < x < 60$ m. The classical scheme at 128^2 captures the depression and deviates only at the fronts.

Neither error mode responds to the parameterisation. The three curves in panel (F) lie within a few centimetres of one another over most of the domain and differ visibly only in the immediate neighbourhood of the bore. What limits a strong-form network on this problem is therefore not a property that the weight spectrum controls.

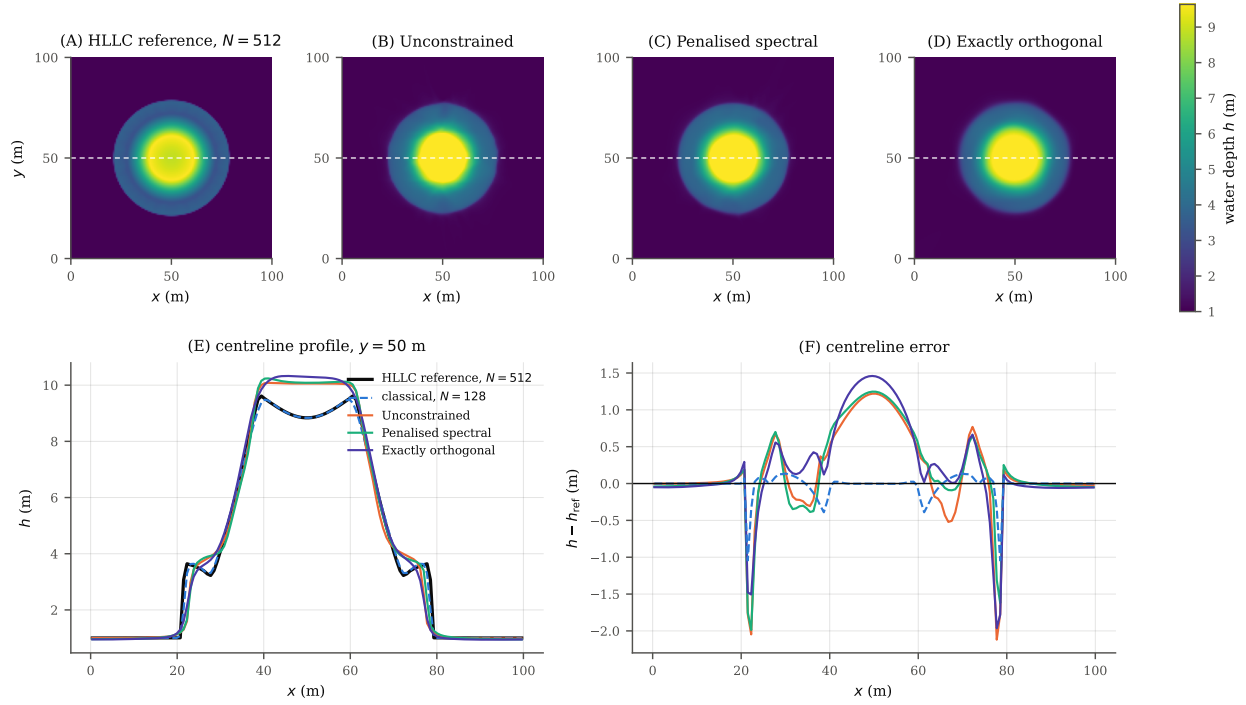


Figure 3: (Color online) Circular dam break over a wet bed at $t = 2$ s, seed 0, on the 128^2 evaluation grid. (A) HLLC reference at $N = 512$; (B)–(D) the unconstrained, penalised spectral and exactly orthogonal networks. The dashed line marks the centreline sampled below. (E) Depth along $y = 50$ m, with a classical HLLC scheme at the evaluation resolution for comparison. (F) The same profiles as deviations from the reference. The networks share two error modes, a displaced and smeared bore and a flattened crest, and the weight parameterisation changes neither.

The two benchmarks that do separate separate against the method. On the Gaussian mound — the only smooth initial condition in the set, and the case closest to the regime the source paper validates in — exact orthogonality reaches $(9.69 \pm 1.8) \times 10^2$ against the unconstrained network’s $(3.83 \pm 0.22) \times 10^2$, and spectrum-only training $(9.16 \pm 0.34) \times 10^2$. This is a factor of 2.5 in the adverse direction, well outside the seed spread, and Section 5.5 places the reference uncertainty on this benchmark at 3.8×10^2 , so the difference is resolvable where smaller ones are not. The penalised parameterisation is intermediate at $(4.69 \pm 0.55) \times 10^2$.

Fig. 4 shows where the additional error lies. All four fields reproduce the collapsing mound and the outward-spreading front, and the signed error fields share one spatial structure: a positive core at the crest, ringed by a negative annulus at the mound edge where the depth gradient is steepest. Constraining U does not move that structure or introduce a new one; it scales it. The ring deepens from panel (E) to panel (G) while its radius and sign pattern are unchanged. Whatever the constraint costs is paid where the solution varies most rapidly, and on this benchmark that is a smooth but steep feature rather than a discontinuity.

On the three-hump case with Manning friction the ordering inverts: spectrum-only training is

Table 4: L^1 depth error at the final output time on the 128^2 evaluation grid, mean $\pm$ s.d. over 5 seeds. Each column carries its own factor. Section 5.5 gives the reference uncertainty for the wet, step and Gaussian columns; it exceeds several of the differences here.

Entry	Circular, wet ($\times 10^3$)	Circular, dry ($\times 10^3$)	Step ($\times 10^3$)	Gaussian mound ($\times 10^2$)	Three humps ($\times 10^2$)
Classical, 128^2	0.391	0.380	0.248	0.064	0.797
Unconstrained	1.602 ± 0.012	1.681 ± 0.076	1.028 ± 0.11	3.83 ± 0.22	8.87 ± 0.68
Fourier features	1.681 ± 0.11	1.838 ± 0.12	1.124 ± 0.068	4.03 ± 0.48	7.96 ± 1.5
Penalised spectral	1.662 ± 0.12	1.602 ± 0.054	1.106 ± 0.047	4.69 ± 0.55	7.49 ± 1.1
Exactly orthogonal	1.725 ± 0.13	1.507 ± 0.14	1.054 ± 0.071	9.69 ± 1.8	9.19 ± 2.3
Spectrum only	2.347 ± 0.11	2.314 ± 0.19	1.126 ± 0.015	9.16 ± 0.34	6.08 ± 1.1
FV-informed	14.81 ± 2.4	9.32 ± 0.54	13.65 ± 2.7	44.0 ± 5.6	8.66 ± 0.22

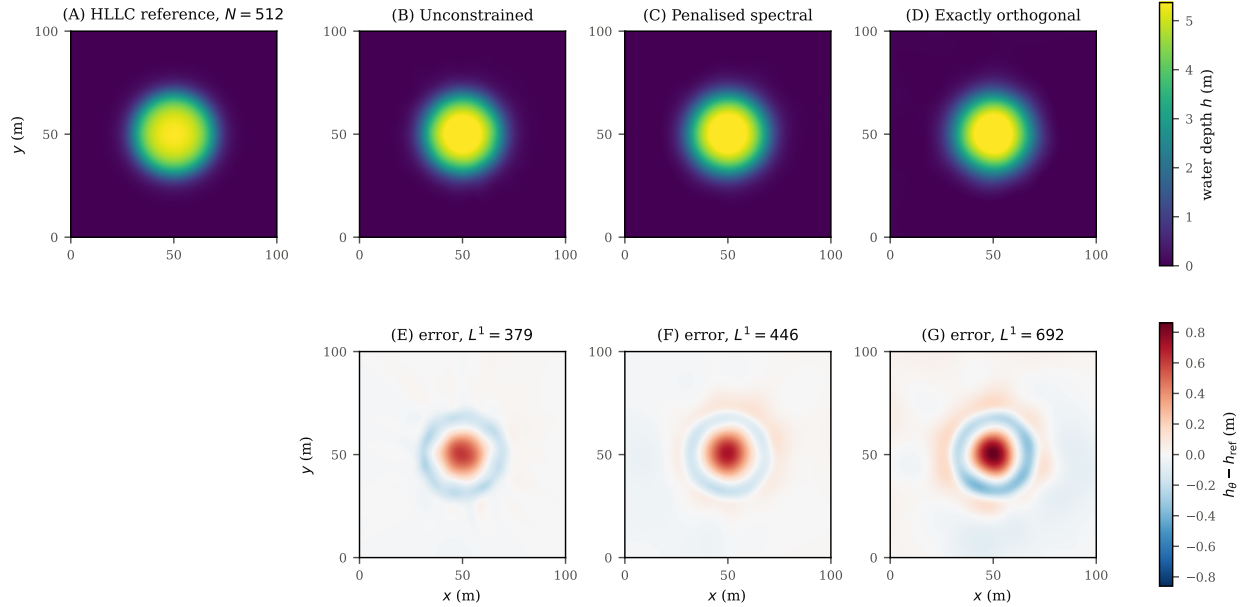


Figure 4: (Color online) Water depth on the Gaussian-mound benchmark at $t = 2$ s, seed 0, on the 128^2 evaluation grid. Top row: (A) the HLLC reference at $N = 512$; (B) the unconstrained network; (C) the penalised spectral parameterisation; (D) exact orthogonality. Bottom row: the corresponding signed error fields (E)–(G) on a common scale, with the L^1 depth error of each. The error is concentrated in a ring at the edge of the mound and at its crest, and grows with the strength of the constraint placed on U while its spatial distribution is unchanged.

the best neural entry at $(6.08 \pm 1.1) \times 10^2$ and exact orthogonality the worst at $(9.19 \pm 2.3) \times 10^2$. We do not read this as support for the spectral family, because every entry on that benchmark loses between 63% and 90% of its initial mass over the 20 s run — against 10^{-3} or below on the two-second cases — so all six models are far outside any regime where a depth error is meaningful. The case is reported for completeness and should not be used to rank methods.

The finite-volume-informed network collapses on the four short cases, matching the behaviour reported for it in [17].

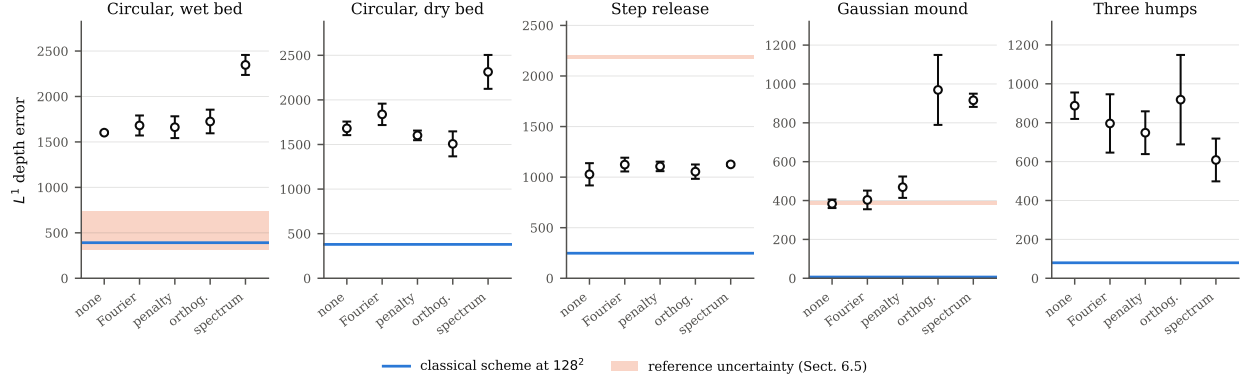


Figure 5: L^1 depth error by benchmark, mean $\pm$ s.d. over 5 seeds; the finite-volume-informed entry is off-scale and omitted. The blue line is a classical scheme at the evaluation resolution. The orange band, where present, spans the disagreement between our reference and two independently implemented schemes on the same initial condition (Section 5.5); on the step release it sits above every neural result, and on the Gaussian mound it is the width of the vanilla network’s own error. Only the Gaussian mound separates the regimes, and it separates them against the constrained ones.

5.3 Sensitivity to the penalty weight

The first of the source paper’s stated limitations is that w_U must be tuned per equation, and no published value exists for shallow-water flow. We bracketed it over three decades on the two circular benchmarks with the corrected Eq. (1) penalty, five seeds per cell (Table 5).

Table 5: L^1 depth error against w_U over three decades, five seeds, using the 2-norm penalty of Eq. (1). The spread across the whole range is smaller than the spread across seeds within any one cell.

w_U	Circular, wet bed	Circular, dry bed
10^{-3}	$(1.578 \pm 0.025) \times 10^3$	$(1.765 \pm 0.032) \times 10^3$
$1/35000 = 2.9 \times 10^{-5}$	$(1.601 \pm 0.12) \times 10^3$	$(1.709 \pm 0.056) \times 10^3$
10^{-6}	$(1.549 \pm 0.069) \times 10^3$	$(1.643 \pm 0.050) \times 10^3$

Three orders of magnitude in w_U move the error by 3% on the wet bed and 7% on the dry bed, both inside the seed-to-seed spread. On these problems the tuning burden the source paper identifies does not exist, because the knob does not do anything. That is a negative answer to their first limitation in this regime rather than a solution to it: there is nothing to adapt. It also says the

penalty is not the active ingredient here, which is consistent with the whole spectral family sitting within 10% of an unmodified network on the same benchmarks.

This sweep supersedes one reading we drew from the earlier, Frobenius-penalty run and now retract. There, the penalised network placed the radial front on the dry bed at 3.5 ± 4.1 m from the reference against the unconstrained network’s 21.2 ± 8.6 m, which we took as a local advantage invisible to the integrated error. Under the corrected penalty the same quantity is 14.8 ± 6.6 m at $w_U = 1/35000$ and between 14.8 and 17.0 m across the three decades, overlapping the unconstrained baseline within one standard deviation. The apparent front-position advantage belonged to the stronger Frobenius penalty, not to the published method, and it does not survive. Integrated depth error was almost unaffected by the same change (1.662×10^3 against 1.601×10^3 on the wet bed), so the norm choice matters for where the shock is placed and not for the depth field as a whole.

5.4 Computational cost

The second stated limitation is the penalty’s training overhead, which the source paper puts at 1.28 to 1.83 times vanilla across its width and collocation sweeps. Table 6 gives wall-clock training time over all 25 runs per entry on one Quadro RTX 8000.

Table 6: Mean training time per run, 25 runs per entry (5 benchmarks $\times$ 5 seeds), identical iteration budgets. The penalty is nearly free here; the penalty-free alternative is the expensive one.

Entry	Mean training time (s)	Relative to unconstrained
Unconstrained	861	1.00 $\times$
Spectrum only	862	1.00 $\times$
Penalised spectral	884	1.03 $\times$
Fourier features	946	1.10 $\times$
Exactly orthogonal	1226	1.42 $\times$
FV-informed	2048	2.38 $\times$

The overhead the source paper reports does not reproduce in this setting: the penalty costs 3%, not 28–83%. One 2-norm of a 128×128 matrix per factorised layer per step is negligible against 20000 collocation points, and we suspect the published figures reflect a smaller problem where the penalty is a larger share of the step, or a different implementation of the norm.

The direction of the comparison matters more than the absolute numbers. Removing the penalty by constraining the factor exactly costs 42%, because the matrix-exponential retraction runs on every forward pass. So on cost grounds the penalty-free route is worse than the penalty, not better, and the motivation for eliminating w_U has to rest on the tuning burden alone — which Section 5.3 shows does not exist on these problems either. Spectrum-only training is free relative to the baseline, as expected from training n parameters per layer instead of n^2 , but it buys nothing in accuracy.

5.5 Uncertainty in the reference solution

Every number in Table 4 is measured against one solver’s output. To find out what that is worth, we compare the reference against an independent implementation: the HLL, Lax–Wendroff and MUSCL–Rusanov runs produced for [17] on a 501^2 node grid for the same initial conditions, interpolated onto the evaluation grid. Those runs store depth only, so the comparison is on depth.

Three regimes appear. On the wet-bed column the reference is vindicated: it agrees with an independent HLL implementation to 3.11×10^2 , tighter than its own 128^2 discretisation error of

Table 7: L^1 depth disagreement between our 512^2 HLLC reference and three independently implemented schemes on the same initial condition, against our own discretisation error and the unconstrained network’s error. Read down a column: where the middle entries exceed the last, the benchmark cannot resolve differences of the size Table 4 reports.

	Circular, wet bed	Gaussian mound	Step release
Our classical at 128^2	3.914×10^2	6.357×10^0	2.476×10^2
vs. their HLL	3.111×10^2	3.763×10^2	2.169×10^3
vs. their MUSCL–Rusanov	7.395×10^2	3.949×10^2	2.208×10^3
vs. their Lax–Wendroff	7.551×10^3	3.435×10^3	9.349×10^3
Unconstrained network	1.602×10^3	$\sim 3.9 \times 10^2$	1.028×10^3

3.91×10^2 , and both are far below the 1.60×10^3 network error. Differences between network variants on that benchmark are real. On the Gaussian mound the two solver families disagree by 3.76×10^2 and the network error is 3.9×10^2 — the same number — while our own discretisation error is sixty times smaller at 6.36. On the step release the solver families differ by 2.2×10^3 , twice the network error. No ranking of network variants on those two benchmarks survives a change of reference solver.

Lax–Wendroff is the outlier throughout, which is expected of Lax–Wendroff with artificial viscosity at a shock and is a statement about that scheme rather than about ours. Nothing here says our reference is wrong; it is the most refined and least diffusive of the four, and on the wet-bed column it is confirmed. What it says is that a comparison table of this kind has a resolution floor that varies by benchmark, and that the floor has to be quoted next to any margin claimed from it. We have not seen this check performed in the PINN-for-conservation-laws literature, where a single fine-grid run is the usual reference.

6 Limitations

Five seeds per cell resolve a factor of two, not the 10% differences that make up most of Table 4; only the Gaussian-mound column separates by more than the spread. Section 5.5 bounds what the comparison can resolve at all and should be read before Table 4, not after it. The three-hump case is reported but not interpretable: every model there loses most of its mass.

The factor-geometry measurement of Table 3 uses 3 seeds on one one-dimensional problem, one architecture and one optimiser, and it is the basis for the paper’s central claim about what the penalty does. It should be repeated on the shallow-water networks, where the per-run factor diagnostics are now recorded but not yet analysed.

The singular-value box of Section 4.3, which is what the refuted prediction points to, is not implemented. Until it is, the claim that the penalty acts by pulling the whole spectrum toward 1 rests on one measurement of D and one failed alternative, both on the advection case.

Two of the five contributions in the project proposal have no section here. The pointwise condition-number field, which is the diagnostic that would explain *where* on the shock the spectral variants fail rather than only that they do, is not built. Neither is the Navier–Stokes leg — the Taylor–Green case the source paper reports and the lid-driven cavity — which is what would connect these results back to the setting [16] validates in. Cost and memory were recorded per run but are not tabulated, so the second stated limitation of the source paper is not yet answered quantitatively.

7 Conclusion

The orthogonality penalty of [16] is load-bearing on their own advection problem and is not doing what the paper says it does. At the published weight the trained factor is rank-deficient rather than orthogonal, and the trained singular values are not the effective weight’s spectrum, yet removing the penalty destroys the accuracy gain and enforcing exact orthogonality recovers only about half of it. Replacing the penalty with a bound on the factor’s largest singular value — the weaker constraint that reading suggests — is the worst of the six regimes we tried. That rules the scale-bound reading out rather than replacing it: the bounded networks match the penalised ones on the orthogonality defect D while missing the effective spectrum by two further orders of magnitude, so D is too coarse a statistic to characterise what the penalty does, and what it does instead remains open.

Moving to shock-dominated shallow-water flow, outside the reach of the source paper’s Theorem 2.1, the picture is flatter and in one place reversed. On the three shock cases no spectral regime separates from an unmodified network by more than about 10% in integrated depth error, and the penalty weight can be moved three decades without measurable effect, so the tuning problem the source paper poses does not arise. On the one smooth case in the set, exact orthogonality and spectrum-only training are a factor of 2.5 *worse* than an unmodified network, by a margin that exceeds the reference uncertainty. Whatever the reparameterisation contributes on smooth one-dimensional targets does not transfer, and constraining the factor hurts most where the target is smoothest.

Establishing any of this required checking the reference against an independent solver family, which on two of three benchmarks disagrees with it by more than the differences under test. We would encourage that check as routine practice: a single fine-grid run is the usual reference in this literature, and its uncertainty is rarely quoted.

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